

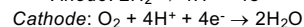
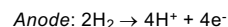
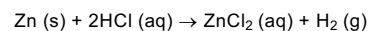


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Propulsion Mechanism

Reactions:



Zinc reacts with hydrochloric acid to produce **hydrogen gas** and zinc chloride in a single displacement, exothermic reaction. The hydrogen gas generated is captured in a sealed system and used as a fuel source for a **hydrogen fuel cell**. To increase the rate of the reaction, copper sulfate (CuSO_4) is used as a catalyst.

The power is generated from a five-stack electric hydrogen fuel cell (Parker TEK-HA-STK-100). The electricity produced from the hydrogen fuel cell will then power the motor to propel the car forward.

Car Design

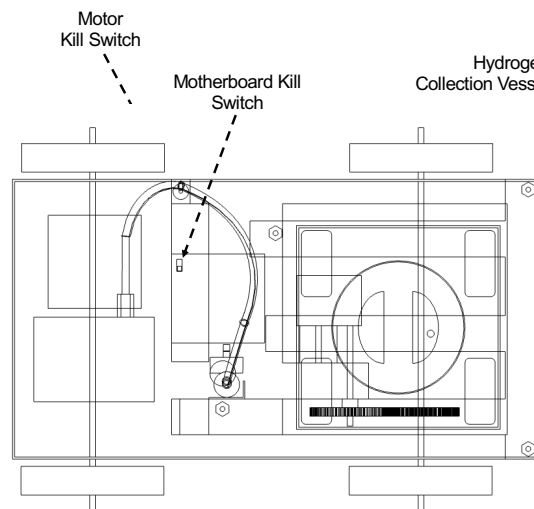


Figure 1: Top-down outline drawing of car base

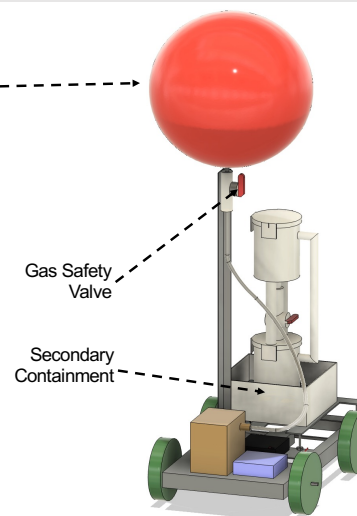


Figure 2: 3-D model of car

Stopping Mechanism

Reactions:

- $\text{IO}_3^- + 3\text{HSO}_3^- \rightarrow \text{I}^- + 3\text{HSO}_4^-$
- $\text{IO}_3^- + 5\text{I}^- + 6\text{H}^+ \rightarrow 3\text{I}_2 + 3\text{H}_2\text{O}$
- $\text{I}_2 + \text{HSO}_3^- + \text{H}_2\text{O} \rightarrow 2\text{I}^- + \text{HSO}_4^- + 2\text{H}^+$
- $2\text{I}_3^- + \text{Starch (I)} \rightarrow \text{Starch-I}_5^- + \text{I}^-$

The stopping mechanism is performed by a **Landolt clock** and an **electro-optical sensor**. The iodate reacts with sodium bisulfite to form iodide, iodate then oxidizes the iodide into iodine. The bisulfite then reduces the iodine into iodide and repeats this until no bisulfite remains in solution. Once this occurs the solution changes to a violet color.

Once the solution turns the dark purple color and covers the photosensor, the sensor will act as a resistor and stop the motor.

Safety Approach

To counteract the risk that hydrogen gas poses, the car's design features a **gas safety valve** to manually block the release of hydrogen into the fuel cell. In addition, the balloon that contains the hydrogen is kept over **18.5 inches** away from any wiring on the vehicle.

The design also has **two deliberate kill switches** built into the motherboard, one for the motor and one for the battery itself. Furthermore, all exposed wiring on the car has plastic covering to ensure complete safety of the operator.

Functional Analysis

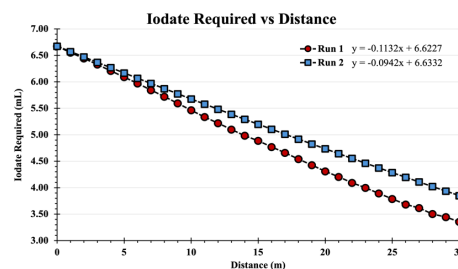


Figure 3: Volume of Iodate vs Distance Plot used to determine reagent quantities

This model is derived from 2 separate relationships of experimental data: *Iodate Volume vs Time to Switch Color* + *Time Motor Runs vs Distance*. The hydrogen fuel cell becomes saturated after the initial run and performs more efficiently, resulting in a faster velocity and shorter Landolt clock period.

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Environmental Impact

Though ZnCl poses a risk to aquatic life, there is the potential to recover the Zinc through electrolysis and reuse it for fuel generation again.

Using CuSO_4 as a catalyst produces solid copper as a byproduct. This solid copper is fully recyclable, eliminating the potential risk posed to aquatic life.

The car's Landolt clock and tubing was constructed using recycled materials. This decreases both the environmental effect and implicit costs of building the car.

